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Retraining the MobileNet Model

Capturing the Data

Performing Inference From the Webcam Feed

Lecture Notes (Optional)

End of Access to Lab Notebooks

Graded Exercise - Rock Paper Scissors

✓ Reading: Exercise Description
10 min

📄 Programming Assignment: Week 4 - Rock Paper Scissors
3h

Course 1 Wrap up

Exercise Description

Rock, Paper, Scissors, Lizard, Spock.

In this week's exercise, you will train your model in the browser using your webcam. You will train your model in exactly the same way as in the Rock, Paper, Scissors example but now you will include the Lizard and Spock hand gestures.

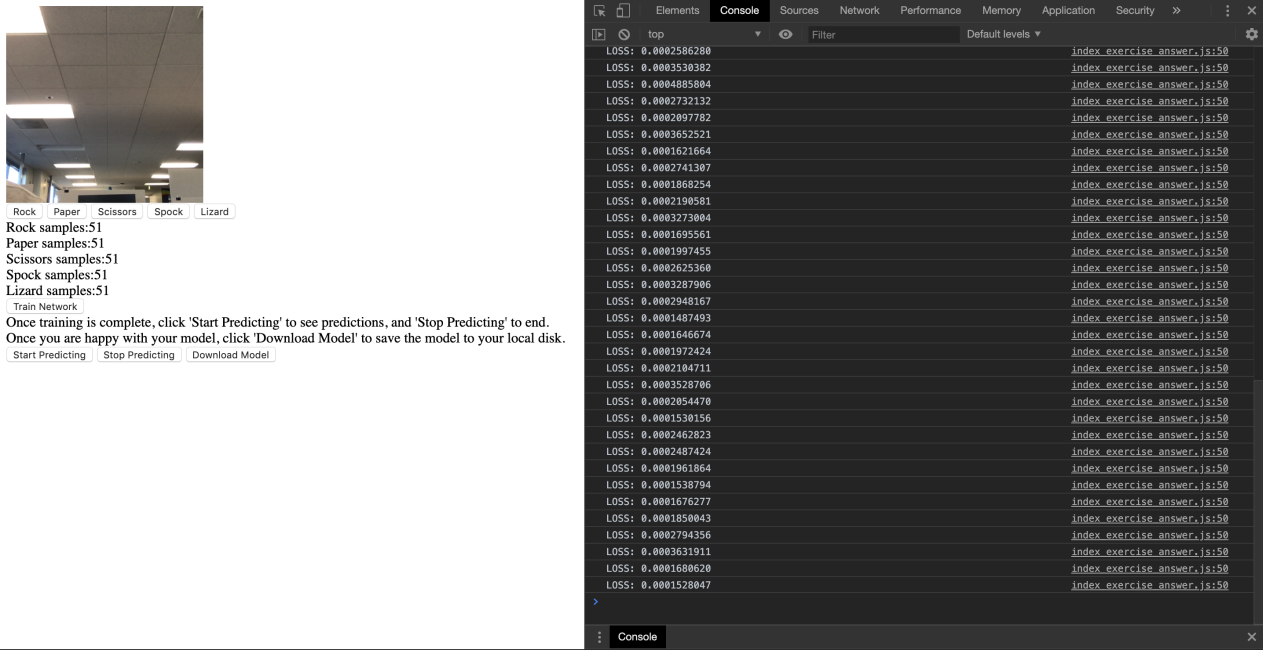
Below are a few tips that can help you get started.

1. To prevent overfitting do not add a lot of dense layers. Using the same layers as in the Rock, Paper, Scissors example should be good enough.

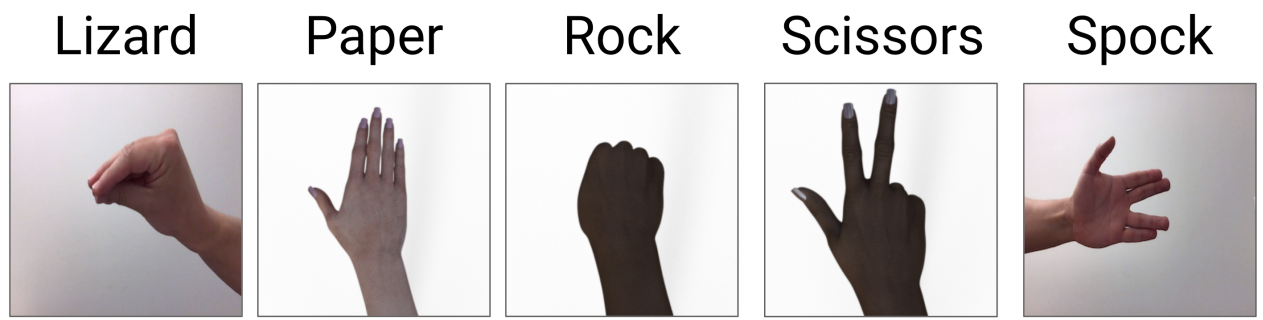
```
15 ▼ async function train() {
16   dataset.js = null;
17   dataset.encodeLabels(5);
18
19   // In the space below create a neural network that can classify hand gestures
20   // corresponding to rock, paper, scissors, lizard, and spock. The first layer
21   // of your network should be a flatten layer that takes as input the output
22   // from the pre-trained MobileNet model. Since we have 5 classes, your output
23   // layer should have 5 units and a softmax activation function. You are free
24   // to use as many hidden layers and neurons as you like.
25   // HINT: Take a look at the Rock-Paper-Scissors example. We also suggest
26   // using ReLu activation functions where applicable.
27 ▼ model = tf.sequential({
28 ▼   layers: [
29
30     // YOUR CODE HERE
31     tf.layers.flatten({inputShape: mobilenet.outputs[0].shape.slice(1)}),
32     tf.layers.dense({ units: 100, activation: 'relu'}),
33     tf.layers.dense({ units: 5, activation: 'softmax'})
34   ]
35 });
```

2. Collecting between 50 to 100 images for each hand gesture should be good enough for most models. However, feel free to experiment and try to train your model with less or more images.

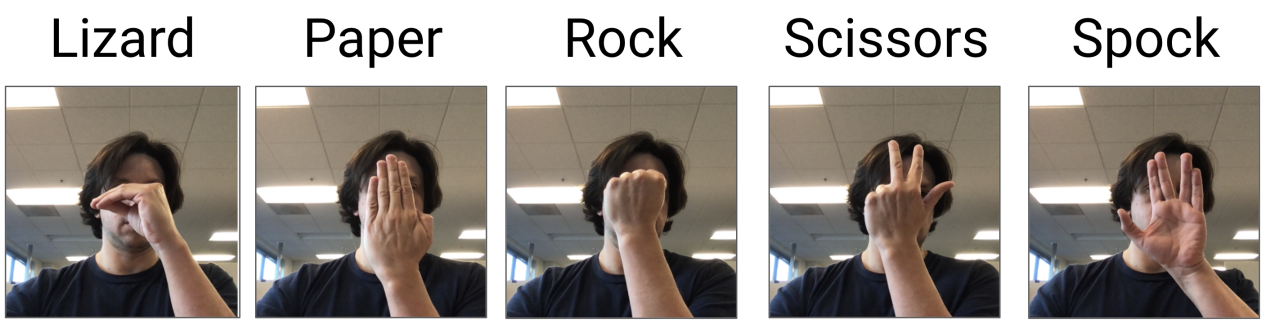
3. **Wait until Training has Finished before you Download the Model.** To make sure training has completed, open the Developer Tools and look at the Console output. When the browser alerts that "Training is Done!" click Ok. After you click Ok, you will see the value of the LOSS being printed in Console. Once the LOSS values stop being printed, you can go ahead and click the "Download Model" button to download the model and its weights.



4. Your model will be graded based on how it performs on our test set. Below are a couple of images from our testing set:



5. To train your model, don't stand too far away from the webcam and it's much better if you do not wear sleeves. Below are some sample training images.



Possible Issues

Because some of the weights of your model are initialized randomly, your model might get stuck in a local minima during training. When this happens your model may not perform optimally and may cause your submission to fail. If you are confident you did everything correctly and your submission didn't pass, try re-training the model and re-submitting it.

Go to next item ✓ Completed

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